Phototaxis Experiment

For this experiment, we will follow the swimming of ***Chlamydomonas Reinhardtii*** **(CR)** under different light conditions. To do so, we will prepare glass slides with the algae trapped between a glass slide and coverslip, and record their motility when only under far-red illumination, and when also illuminated by a green light on one side.

Under green illumination, the movement of the algae should present a change towards or away from the source of light, this phenomenon is called **Phototaxis**.

Using Python library **TrackPy** (soft-matter.github.io/trackpy), we will track the movement of the algae on the recordings made to compare and quantify their behaviour under different green light intensities, and in between periods of illumination.

The camera used is the IDS UI-3370CP-NIR-GL (datasheet). The associated software is **IDS peak** (download here).

Your computer must be restarted after installing IDS peak before you can use it!

1 Outline of the Session

1. Presentation of the session activity and aims.
2. Calibrate image with a stage micrometer, and prepare several glass slides with algae (kept in the dark).
3. Prepare sample 1 and use it to select parameters for the movies.
4. Prepare sample 2 (if needed) and use it to take a movie while turning the green light on and off, such as: (1) far-red only, (2) with “weak” green illumination, (3) far-red only, (4) with “weak” green illumination, (5) far-red only, (6) with “strong” green illumination, (7) far-red only.
5. Transfer and analyse the movies.

Note: For each step, it might be useful to read the whole set of instructions before you start to work.

2 Prepare Samples and Image Calibration

We will make simple chambers to prepare the samples. A first sample will also help us find optimal recording parameters (background illumination, framerate, camera exposure, cell concentration).

- Open the camera software **IDS peak** (already installed on the recording computer).
- To calibrate the images, place the **Stage Micrometer**. Take a snapshot image of the micrometer. Use **Fiji** to measure the length in pixels of a known distance and get the pixel-to-micron conversion.

  Write down this pixel/µm conversion value — you will need it for tracking.

We will now prepare sample cells with a microscope slide and a coverslip, sandwiching double-sided sticky tape or thinly cut parafilm.

Prepare at least 2 samples!

- Wear gloves to avoid greasing the glass. Cut two thin bands of double-sided sticky tape (or parafilm) and stick onto a microscope slide with a free space in between. Top with a coverslip. If using parafilm, heat the slide to melt it, apply gentle pressure on the coverslip over the bands, and wait for it to cool down.
- Put a cone on the pipettor, cut the tip slightly wider, and fill the chamber with the *Chlamydomonas* cell suspension (normally less than 20 µl). Stop before the liquid reaches the other end of the coverslip. Absorb excess liquid from the filling side with a kimtech wipe.
- Seal both ends of the chamber with grease (avoid grease contact with the culture).
- Place one slide on the microscope.
- Adjust background illumination, focus, and camera parameters. Balance: good contrast without saturating the camera, a suitable framerate to follow algae swimming, and convenient cell concentration.

  Write down these parameters — they will be needed for tracking.
- Discuss whether you need to dilute the cell suspension to track cells and determine their instantaneous velocity.
- If needed, dilute the suspension by the required amount using cell culture medium (TAP). This suspension will be used in the following step.
- Turn on the green light and verify the CR swimming reaction. Try at lowest intensity first, then highest.
- Set the LED back to lowest intensity, then turn it off.

3 Record Swimming of Algae

We want to record movies of cell behaviour with/without an external green-light stimulus, then analyse them to quantify light-induced changes in behaviour.

- Cover the setup with the black sheets to prevent unwanted light stimuli.
- Make sure the green LED is **OFF**. Leave the chamber under far-red illumination only for at least **5 minutes**. Place the other chamber in the dark (e.g. under aluminium foil).
- Bring the slide into focus. Verify all recording parameters, including that frames are saved as `.tif` files and that the output directory is created.
- Prepare a chronometer. Read the full timeline below before starting.
- Start recording and the chronometer simultaneously, then follow the timeline:

$t = 0$ s	**Start recording**	
$t = 30$ s	**Turn ON green light**	low intensity
$t = 60$ s	**Turn OFF green light**	
$t = 90$ s	**Turn ON green light**	low intensity
$t = 120$ s	**Turn OFF green light**	
$t = 150$ s	**Turn ON green light**	high intensity
$t = 180$ s	**Turn OFF green light**	
$t = 215$ s	**Stop recording**	

- The experimental part is done. You can dispose of the sample and cells.

4 Analyse Movies

Everyone will now use their laptop to analyse their movie using Python and the TrackPy library. Tracking follows two steps:

- **Featuring:** finding features in each frame (coordinates, integrated intensity, size).
- **Tracking:** assigning consistent labels to features across frames to reconstruct trajectories.

More details are available on the TrackPy GitHub.

The required code is in the online folder. The analysis follows this pipeline:

```
Load images → Build background → Process frames →
Test particle detection → Filter noise → Run featuring →
Link trajectories → Compute kinematics → Analyse by lighting condition
```

1. Load images.
2. Calculate the mean frame from the original frames and generate median-subtracted frames. These will be used for tracking. Subtracting the mean removes background and stuck features, highlighting moving ones.
3. Process frames to optimise subsequent particle detection.
4. Test particle featuring.
5. Filter noise.
6. Run featuring on the whole movie.
7. Link trajectories and compute kinematics.
8. Analyse the distribution of feature speeds and velocity orientations.

The aim of the data analysis is to compare cell behaviour across experiments. Potentially interesting questions include (but are not limited to!):

- Does the speed distribution change when the green light is turned ON?
- Do the speed distributions in the control movies differ? In other words, is there any sort of memory of the previous green-light exposure history?
- How does the spread in cell swimming orientation change during phototaxis? Is there any evidence that it could depend on light intensity?